Hydrocortisone (systemic): Drug information - UpToDate

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Contributor Disclosures

For additional information see "Hyԁrocortisone (systemic): Patient drug information" and "Hyԁrocortisone (systemic): Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Alkindi Sprinkle;
- Cortef;
- Khindivi;
- Solu-CORTEF

Brand Names: Canada

- AURO-Hyԁrocortisone;
- Cortef;
- Solu-CORTEF

Pharmacologic Category

- Corticosteroid, Systemic

Dosing: Adult

Dosage guidance:

Dosing: Individualize dosing and use the minimum effective dose.

Adrenal insufficiency, adrenal crisis

Adrenal insufficiency, adrenal crisis:

Note: Appropriate fluid resuscitation is also required (Ref).

Initial dosing: **IV:** 100 mg IV bolus given immediately, followed by 50 mg every 6 hours or 200 mg/24 hours as a continuous IV infusion for the first 24 hours (Ref).

Subsequent management: If clinical status has improved after 24 hours, may begin gradually tapering the dose (eg, to 50 mg IV every 12 hours). Once patient is stable, resume or begin chronic maintenance oral regimen (Ref). In patients with mineralocorticoid deficiency (eg, primary adrenal insufficiency), use in combination with fludrocortisone once hyԁrocortisone dose is <40 mg/day (Ref).

Adrenal insufficiency, chronic

Adrenal insufficiency, chronic:

Note: Use in combination with fludrocortisone in patients with mineralocorticoid deficiency (eg, primary adrenal insufficiency) (Ref).

Maintenance dosing: **Oral:** Initial: 15 to 25 mg/day in 2 or 3 divided doses (Ref). May adjust daily dose by 2.5 to 5 mg/day if needed based on signs and symptoms of under- or over-replacement; use the lowest effective dose (Ref). Daily dose may be divided as follows:

2-dose regimen: Administer two-thirds of the daily dose in the morning upon awakening and one-third of the daily dose two hours after lunch (Ref).

3-dose regimen: Administer the largest dose in the morning, followed by decreasing doses at lunch and in the afternoon no later than 4 to 6 hours before bedtime (eg, 10 mg at 7 am, 5 mg at noon, and 2.5 mg at 4 pm) (Ref).

Stress dosing: **Note:** For use in patients with chronic adrenal insufficiency (AI) and acute physiologic stressors (eg, illness, surgery) to prevent development of adrenal crisis. Patients without known chronic AI with recent or current chronic glucocorticoid use at supraphysiologic doses (eg, >20 mg/day hydrocortisone) may have hypothalamic-pituitary-adrenal axis suppression and may require stress dosing; individualize treatment decisions in these patients (Ref).

Acute physiologic stress/illness:

Febrile illness: **Oral:** Double the chronic maintenance dose for fever 38°C (100.4°F) to 39°C (102.2°F) or triple the chronic maintenance oral dose for fever >39°C (>102.2°F) or persistent nausea. Continue higher dose for 3 days, then return to baseline dose if fever has resolved. If fever has not resolved by day 4, further evaluation (with continued use of higher steroid doses if needed) is required (Ref).

Patients unable to tolerate oral medication (eg, due to vomiting or diarrhea): **IV, SUBQ, IM:** 100 mg given early in course of illness (IM, SUBQ dosing may be self-administered); may repeat dose after 6 to 12 hours under medical supervision (Ref).

Illness requiring hospitalization (eg, community acquired pneumonia): **IV, Oral, IM:** 50 to 75 mg/day in 2 or 3 divided doses. Taper and return to baseline dose within 2 to 3 days following improvement of underlying condition (Ref). **Note:** For critical illness requiring intensive care, follow dosing recommendations for adrenal crisis (Ref).

Labor/Delivery: Dosing is individualized; an example regimen is as follows:

IV, IM: 100 mg IV or IM at the onset of active labor (or IV immediately before anesthesia in patients undergoing caesarean section), followed by 50 mg IV or IM every 6 hours **or** 200 mg/24 hours as a continuous IV infusion until delivery. Taper dose over 1 to 2 days after delivery before resuming baseline dose (Ref).

Surgical stress:

Minor surgical stress (eg, hernia repair, procedures with local anesthetic): **IV, Oral, IM:** Give an additional 25 mg supplemental dose on the day of surgery (Ref).

Moderate surgical stress (eg, joint replacement, cholecystectomy): **IV, IM:** 50 to 75 mg/day (eg, 25 mg every 8 to 12 hours) through postoperative day 1, then resume baseline oral regimen on postoperative day 2 if clinical course is uncomplicated (Ref).

Major surgical stress (eg, major bowel surgery, cardiac bypass, cesarean delivery): **IV, IM:** 100 mg IV immediately before anesthesia, followed by 50 mg IV or IM every 6 hours **or** 200 mg/24 hours as a continuous IV infusion through postoperative day 2 or 3. Then, taper to baseline oral regimen (eg, reduce dose by 50% per day) starting on postoperative day 3 or 4 if clinical course is uncomplicated (Ref).

Adrenal insufficiency due to classic congenital adrenal hyperplasia

Adrenal insufficiency due to classic congenital adrenal hyperplasia:

Note: For patients who have difficulty adhering to multiple daily dosing regimens, long-acting glucocorticoids (eg, prednisolone) may be preferred (Ref). In patients with mineralocorticoid deficiency, use in combination with fludrocortisone (Ref).

Oral: 15 to 30 mg/day in 2 to 3 divided doses (Ref). Daily dose may be divided as follows:

3-dose regimen (preferred): Administer the largest dose in the morning upon awakening, followed by decreasing doses at lunch and in the afternoon no later than 4 to 6 hours before bedtime (eg, 15 mg at 7 am, 5 mg at noon, and 2.5 mg at 4 pm) (Ref).

2-dose regimen: Administer two-thirds of the daily dose in the morning upon awakening and one-third of the daily dose two hours after lunch (Ref).

Dosage adjustment: May adjust daily dose by 2.5 to 5 mg/day if needed based on signs and symptoms of under- or over-replacement and degree of androgen suppression; use the lowest effective dose (Ref).

Note: If androgen suppression cannot be achieved with doses ≤30 to 35 mg/day, or if sleep impairment or signs of excessive glucocorticoid replacement occur, may consider combination therapy with a long-acting glucocorticoid (eg, methylprednisolone, prednisolone) at bedtime (Ref).

Stress dosing: Patients undergoing physiologic stress (eg, febrile illness, surgery, labor/delivery) require increased glucocorticoid doses to prevent adrenal crisis. Individualize treatment decisions; refer to dosing for “Adrenal insufficiency, chronic” for stress dosing instructions (Ref).

Adrenal insufficiency associated with immune checkpoint inhibitor therapy

Adrenal insufficiency associated with immune checkpoint inhibitor therapy: Note: For oral hydrocortisone, usually $^{2}/_{3}$ of the dose is administered in the morning and $^{1}/_{3}$ of the dose is administered in the early afternoon (Ref).

Initial:

Grade 1: **Oral:** 15 to 20 mg/day in divided doses; if there are residual symptoms, titrate up to a maximum of 30 mg/day (Ref).

Grade 2: **Oral:** 30 to 50 mg/day in divided doses (Ref).

Grade 3 or 4: **IV:** 50 to 100 mg every 6 to 8 hours; taper down to oral maintenance corticosteroids over 5 to 7 days (Ref).

Maintenance: Oral: 15 to 20 mg/day in divided doses (Ref).

Anti-inflammatory or immunosuppressive

Anti-inflammatory or immunosuppressive (alternative agent):

Note: In patients receiving chronic glucocorticoids (eg, ≥3 to 4 weeks) at supraphysiologic doses (>20 mg/day hydrocortisone) **for indications other than adrenal insufficiency**, risk of hypothalamic-pituitary-adrenal axis suppression is increased. If glucocorticoid discontinuation is indicated, reduce dose gradually and monitor for signs of adrenal insufficiency. Higher doses may be needed during acute illness or surgery (Ref).

IM, IV: Initial: 100 to 500 mg/dose at intervals of 2, 4, or 6 hours.

Oral: Initial: 20 to 240 mg/day.

Asthma, acute exacerbation

Asthma, acute exacerbation (alternative agent) (off-label dose): Note: Alternative to other corticosteroids in moderate to severe exacerbations or in patients who do not respond promptly and completely to short-acting beta-agonists; administer within 1 hour of presentation to emergency department (Ref).

Oral: 200 mg in divided doses for 5 to 7 days (Ref).

COVID-19, hospitalized patients

COVID-19, hospitalized patients (alternative agent) (off-label use):

Note: Hydrocortisone is recommended for treatment of COVID-19 in hospitalized patients requiring supplemental oxygen or ventilatory support when dexamethasone is not available or there are specific indications for hydrocortisone. Dosing is extrapolated from a study that used dexamethasone; the equivalent dose of hydrocortisone (or other glucocorticoid) may be substituted if necessary (Ref).

IV, Oral: 50 mg every 8 hours for up to 10 days (or until discharge if sooner) as part of an appropriate combination regimen (Ref).

Deceased organ donor management

Deceased organ donor management (hormonal replacement therapy) (off-label use):

Note: Data supporting benefit are conflicting; refer to institutional protocols (Ref). If given, administer as soon as possible after brain death is declared, but **after** blood has been collected for tissue typing (Ref).

IV: Example regimens include: 300 mg once, followed by 100 mg every 8 hours **or** 50 mg bolus, followed by a 10 mg/hour infusion (Ref).

Hypophysitis associated with immune checkpoint inhibitor therapy

Hypophysitis associated with immune checkpoint inhibitor therapy:

Initial:

Grade 1: **Oral:** 15 to 20 mg/day in divided doses; if there are residual symptoms, titrate up to a maximum of 30 mg/day (Ref).

Grade 2: **Oral:** 30 to 50 mg/day in divided doses (Ref).

Grade 3 or 4: **IV:** 50 to 100 mg every 6 to 8 hours; taper down to oral maintenance corticosteroids over 5 to 7 days (Ref).

Maintenance: **Oral:** 15 to 20 mg/day in divided doses (Ref) **or** 10 to 12 mg/m^2/day (Ref).

Iodinated contrast media allergic-like reaction, prevention

Iodinated contrast media allergic-like reaction, prevention: **Note:** Generally reserved for patients with a prior allergic-like or unknown-type iodinated contrast reaction who will be receiving another iodinated contrast agent. Nonurgent premedication with an oral corticosteroid is generally preferred when contrast administration is scheduled to begin in ≥12 hours; however, IV hydrocortisone may be considered in patients unable to take oral medications. Consider an urgent (accelerated) regimen with an IV corticosteroid for patients requiring contrast in <12 hours (Ref).

Nonurgent regimen (alternative agent): **IV:** 200 mg administered 13 hours, 7 hours, and 1 hour before contrast medium administration in combination with oral or IV diphenhydramine (Ref).

Urgent (accelerated) regimen: **IV:** 200 mg immediately, then 200 mg every 4 hours until contrast medium administration in combination with oral or IV diphenhydramine. In emergent situations (ie, <4 to 5 hours until contrast is required), may consider hydrocortisone 200 mg in combination with IV diphenhydramine 1 hour prior to contrast (Ref).

Pneumonia, community acquired, severe

Pneumonia, community acquired, severe (adjunctive agent) (off-label use):

IV: 200 mg bolus, followed by 10 mg/hour continuous infusion for 7 days (Ref) **or** 200 mg/day as a continuous infusion for 4 to 7 days (based on clinical improvement), then taper for a total duration of 8 to 14 days with discontinuation upon ICU discharge (Ref).

Septic shock

Septic shock (off-label use):

Note: For use in patients with ongoing vasopressor requirements; may administer with or without fludrocortisone (Ref).

IV: 200 to 300 mg/day continuous infusion or in divided doses for 5 to 7 days or until ICU discharge (Ref).

Thyroid storm

Thyroid storm (off-label use): IV: 300 mg loading dose, followed by 100 mg every 8 hours; use in combination with other appropriate agents (Ref). The optimal duration of therapy is unknown; taper and discontinue therapy as clinical status improves (Ref).

Ulcerative colitis, acute, remission induction

Ulcerative colitis, acute (severe), remission induction (off- label dose): IV: 100 mg every 6 to 8 hours (Ref). Some experts recommend not exceeding 300 mg/day to decrease the risk of colectomy (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Note: The pharmacokinetics and pharmacodynamics of hyدrocortisone in kidney impairment are not well understood (Ref). Cortisone metabolism and excretion may be impaired with kidney dysfunction (cortisone in its free form and its breakdown products are excreted in the urine) (Ref). The clinical implications of these pharmacokinetic alterations are unclear.

Altered kidney function: Mild to severe impairment: No dosage adjustment necessary (Ref).

Hemodialysis, intermittent (thrice weekly): No supplemental dose or dosage adjustment necessary (Ref).

Peritoneal dialysis: No dosage adjustment necessary (Ref).

CRRT: No dosage adjustment necessary (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): No dosage adjustment necessary (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling; use with caution.

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2, and 3 obesity (BMI ≥30 kg/m^2):

IV, IM, Oral: No dosage adjustment necessary (expert opinion). Refer to adult dosing for indication-specific doses.

Rationale for recommendations: Corticosteroids are lipophilic compounds; however, the reported pharmacokinetic variability due to obesity is limited and inconsistent. One case report described cortisol pharmacokinetics following hyدrocortisone administration pre- and post-bariatric surgery. Following bariatric surgery, both cortisol V_d and clearance were reduced (Ref). The clinical relevance of these pharmacokinetic changes requires further study.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Hyدrocortisone (systemic): Pediatric drug information")

Dosage guidance:

Dosing: Adjust dose depending upon condition being treated and response of patient. The lowest possible dose should be used to control the condition; when dose reduction is possible, the dose should be reduced gradually. In life-threatening situations, parenteral doses larger than the oral dose may be needed.

Dosage form information: Due to inactive ingredients (eg, polyethylene glycol 400, propylene glycol, and glycerin), the commercially available 1 mg/mL solution (Khindivi) is not recommended for use in infants and children <5 years of age or for stress dosing. Use of oral sprinkles or an extemporaneously compounded solution may be considered.

Safety: Extemporaneously compounded oral suspensions are available in multiple concentrations (eg, 2 mg/mL, 2.5 mg/mL) and commercially available oral solution is 1 mg/mL; caution should be taken to verify and avoid confusion between the different concentrations; dose should be clearly presented as mg. Use caution when changing between dosage forms (eg, from compounded suspension or crushed tablet to sprinkles or oral solution) and monitor closely for potential differences in exposure; dosage adjustment may be necessary.

Adrenal insufficiency, adrenal crisis

Adrenal insufficiency, adrenal crisis:

Note: Dosage regimens variable; some experts recommend using weight-directed dosing for first dose (Ref):

Weight-directed: Limited data available: Infants, Children, and Adolescents: IM, IV (preferred), Intraosseous: Initial: 2 to 3 mg/kg once; maximum dose: 100 mg/dose; then for infants: 1 to 5 mg/kg/dose every 6 hours; for children and adolescents, see BSA- or age-directed dosing (Ref).

BSA-directed dosing: Limited data available: Infants, Children, and Adolescents: IM, IV (preferred), Intraosseous: Initial: 50 to 100 mg/m^2 once followed by 50 to 100 mg/m^2/day in 4 divided doses (Ref).

Age-directed dosing (fixed dosing): Limited data available (Ref):

Infants: IM, IV (preferred), Intraosseous: 10 to 25 mg once followed by 10 to 25 mg/day in divided doses every 6 hours for 24 hours then subsequent dose reductions and rate determined by patient response.

Children <5 years (eg, young children): IM, IV (preferred), Intraosseous: 25 to 50 mg once followed by 25 to 50 mg/day in divided doses every 6 hours for 24 hours then subsequent dose reductions and rate determined by patient response.

Children ≥5 years (eg, older children): IM, IV (preferred), Intraosseous: 50 to 100 mg once followed by 50 mg/day in divided doses every 6 hours for 24 hours then subsequent dose reductions and rate determined by patient response.

Adolescents: IM, IV (preferred), Intraosseous: 100 mg once followed by 100 mg/day in divided doses every 6 hours for 24 hours then subsequent dose reduction and rate determined by patient response.

Adrenal insufficiency, replacement

Adrenal insufficiency, replacement:

Note: When using the oral sprinkles (Alkindi) or oral solution (Khindivi), round dose to nearest 0.5 or 1 mg. To replicate diurnal variation, the highest doses are typically administered in the morning and midday dose with the lower dose in the evening (Ref). Patients with confirmed aldosterone deficiency should receive fludrocortisone (children and adolescents) along with sodium chloride supplements if needed (infants) (Ref).

Sprinkle (Alkindi), tablets: Infants, Children, and Adolescents: Oral: 8 to 10 mg/m^2/day in 3 divided doses; older patients may be able to transition to twice-daily dosing. Higher doses up to 12 mg/m^2/day may be required in patients with primary adrenal insufficiency; lower doses may be sufficient for patients with secondary adrenal insufficiency (Ref).

Oral solution 1 mg/mL (Khindivi): Children ≥5 years and Adolescents: Oral: 8 to 10 mg/m^2/day in 3 divided doses; older patients may be able to transition to twice-daily dosing. Higher doses up to 12 mg/m^2/day may be required in patients with primary adrenal insufficiency; lower doses may be sufficient for patients with secondary adrenal insufficiency (Ref). **Note:** Do not use commercially available oral solution (Khindivi) for stress dosing; if stress dosing is needed, switch patient to different hydrocortisone product.

Anti-inflammatory or immunosuppressive

Anti-inflammatory or immunosuppressive:

Note: Dosing range variable; individualize dose for disease state and patient response.

Infants and Children:

Oral: 2.5 to 10 mg/kg/day or 75 to 300 mg/m^2/day divided every 6 to 8 hours (Ref).

IM, IV:

Manufacturer's labeling: IM, IV: Initial: 0.56 to 8 mg/kg/day or 20 to 240 mg/m^2/day in 3 or 4 divided doses.

Alternate dosing: Limited data available: IM, IV: 1 to 5 mg/kg/day or 30 to 150 mg/m^2/day divided every 12 to 24 hours (Ref).

Adolescents: Oral, IM, IV, SUBQ: 15 to 240 mg every 12 hours (Ref).

Congenital adrenal hyperplasia, chronic

Congenital adrenal hyperplasia (CAH), chronic:

Note: Administer morning dose as early as possible. Individualize dose by monitoring growth, hormone levels, and bone age; mineralocorticoid (eg, fludrocortisone) and sodium supplement may be required in salt losers. When using the oral sprinkles (Alkindi) or oral solution (Khindivi), round dose to nearest 0.5 or 1 mg; when using regular oral tablets, the smallest available dose is 2.5 mg ($^1/_2$ of 5 mg tablet). See "Stress Dosing; Supplemental" for management of patients with CAH during times of physiological stress.

BSA-directed dosing:

Tablets, sprinkles: Infants, Children, and Adolescents: Oral: Initial: 8 to 15 mg/m^2/day in 3 divided doses; usual range: 10 to 15 mg/m^2/day; older patients may be able to transition to twice-daily dosing (Ref).

Oral solution 1 mg/mL (Khindivi): Children ≥5 years and Adolescents: Oral: Initial: 8 to 10 mg/m^2/day in 3 divided doses; older patients may be able to transition to twice-daily dosing. Higher doses may be needed for some patients; usual range: 10 to 15 mg/m^2/day (Ref). **Note:** Do not use commercially available oral solution (Khindivi) for stress dosing; if stress dosing is needed, switch patient to different hydrocortisone product.

Fixed dosing (Ref):

Infants: Oral: Tablets: Usual requirement: 2.5 to 5 mg/**dose** 3 times daily.

Children: Oral: Tablets: Usual requirement: 5 to 10 mg/**dose** 3 times daily.

Adolescents (fully grown): Oral: Tablets: Usual requirement: 15 to 25 mg/day divided in 2 to 3 daily doses.

Stress dosing; supplemental

Stress dosing; supplemental: Limited data available; dosage regimens variable:

Note: Dosing based on the level of physiological stress related to condition; dose should be individualized based on patient and continued until resolution of stressful condition (usually 24 to 48 hours). Typically, supplementation for emotional or minimal physiological stress conditions or prior to exercise is not necessary. Dosing is generally 2 to 3 times physiologic replacement level (Ref).

Commercially available solution (Khindivi) is not approved for use for stress dosing due to risk of toxicity from inactive ingredients (eg, polyethylene glycol 400, propylene glycol, and glycerin) at the higher doses.

Infants, Children, and Adolescents:

BSA-directed dosing (Ref): Oral, IM, IV:

Mild to moderate stress: 20 to 50 mg/m^2/day divided into 3 or 4 doses; doses on the lower end of the range (20 to 30 mg/m^2/day) may be divided twice daily.

Major stress or surgery: 100 mg/m^2/day in divided doses every 6 hours.

Planned surgery: Pre-anesthesia of 50 mg/m^2 IV or IM administered 30 to 60 minutes prior to surgery followed by second dose of 50 mg/m^2 as a continuous IV infusion or in divided doses every 6 hours for at least 24 hours.

Age-directed for moderate stress in patients with congenital adrenal hyperplasia (Ref):

Infants and preschool children: IV: Initial dose: 25 mg once, followed by 6.25 mg every 6 hours; resume patient's standard maintenance dose once stable.

School-age children: IV: Initial dose: 50 mg once, followed by 12.5 mg every 6 hours; resume patient's standard maintenance dose once stable.

Adolescents: IV: Initial dose: 100 mg once, followed by 25 mg every 6 hours; resume patient's standard maintenance dose once stable.

Septic shock, fluid and catecholamine-refractory with suspected/proven adrenal insufficiency

Septic shock, fluid and catecholamine-refractory with suspected/proven adrenal insufficiency: Limited data available:

BSA-directed dosing : Infants, Children, and Adolescents: IV: 50 to 100 mg/m^2/day. **Note:** The maximum adult daily dose is 200 mg/**day** (~100 mg/m^2/**day**) (Ref). In some cases, doses may be titrated up to 50 mg/**kg**/day as a continuous IV infusion, if necessary, for shock reversal in the short term; however, efficacy data variable with the higher doses (Ref).

Weight- directed dosing (Ref): **Note:** Use if BSA not available:

Infants, Children, and Adolescents: IV, Intraosseous: 2 mg/kg as a single bolus dose; maximum dose: 100 mg/dose.

Age- directed dosing (Ref): **Note:** Use if BSA and weight are not available:

Infants and Children <3 years: IV, Intraosseous: 25 mg as a single bolus dose.

Children ≥3 to <12 years: IV, Intraosseous: 50 mg as single bolus dose.

Children ≥12 years and Adolescents: IV, Intraosseous: 100 mg as a single bolus dose.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; use with caution.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; use with caution.

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified.

Frequency not defined:

Cardiovascular: Bradycardia, cardiac arrhythmia, cardiomegaly, circulatory shock, embolism (fat), heart failure, hypertension, hypertrophic cardiomyopathy (premature infants), syncope, tachycardia, thromboembolism, thrombophlebitis, vasculitis

Dermatologic: Acne vulgaris, allergic dermatitis, atrophic striae, burning sensation of skin (especially in the perineal area after IV injection), desquamation, diaphoresis, ecchymoses, epidermal thinning, erythema of skin, facial erythema, fragile skin, hyperpigmentation, hypertrichosis, hypopigmentation, inadvertent suppression of skin test reaction, skin atrophy (including cutaneous atrophy, subcutaneous atrophy), skin edema, skin rash, thinning hair, urticaria, xeroderma

Endocrine & metabolic: Adrenocortical insufficiency (secondary unresponsiveness, particularly during trauma, surgery, or illness), Cushing syndrome, cushingoid appearance (fat accumulation in cheeks and temporal fossae), decreased glucose tolerance, decreased serum potassium, fluid retention, growth suppression, hirsutism, HPA-axis suppression, hyperglycemia (including increased requirements for insulin or oral hypoglycemic agents in diabetes mellitus), hypokalemic alkalosis, menstrual disease, negative nitrogen balance (due to protein catabolism), pituitary insufficiency (secondary unresponsiveness, particularly during trauma, surgery, or illness), prediabetes, protein catabolism, sodium retention, weight gain

Gastrointestinal: Abdominal distention, gastrointestinal perforation (small and large intestine, particularly in patients with inflammatory bowel disease), hiccups, impaired intestinal carbohydrate absorption, increased appetite, nausea, pancreatitis, peptic ulcer (with possible perforation and hemorrhage), ulcerative esophagitis

Genitourinary: Asthenospermia, glycosuria, oligospermia

Hematologic & oncologic: Leukocytosis, petechia

Hepatic: Hepatomegaly, increased liver enzymes (usually reversible on discontinuation)

Hypersensitivity: Hypersensitivity reaction (including anaphylaxis, angioedema, nonimmune anaphylaxis)

Infection: Increased susceptibility to infection, sterile abscess

Local: Postinjection flare (intra-articular use)

Nervous system: Depression, emotional lability, euphoria, headache, increased intracranial pressure (with papilledema; usually following discontinuation), insomnia, malaise, myasthenia, neuritis, neuropathy, paresthesia, personality changes, psychic disorder, seizure, tingling of skin (especially in the perineal area after IV injection), vertigo

Neuromuscular & skeletal: Amyotrophy, aseptic necrosis of femoral head, aseptic necrosis of humeral head, Charcot arthropathy, lipodystrophy, osteoporosis, pathological fracture (long bones), rupture of tendon (particularly rupture of Achilles tendon), steroid myopathy, vertebral compression fracture

Ophthalmic: Blindness (periocular injection), exophthalmos, glaucoma, increased intraocular pressure, retinopathy (central serous chorioretinopathy), subcapsular posterior cataract

Respiratory: Pulmonary edema

Miscellaneous: Wound healing impairment

Contraindications

Hypersensitivity to hydrocortisone or any component of the formulation; systemic fungal infections; use in premature infants (formulations containing benzyl alcohol only); idiopathic thrombocytopenia purpura (IM administration only); intrathecal administration; live or live, attenuated virus vaccines (with immunosuppressive doses of corticosteroids).

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Canadian labeling: Additional contraindications (not in US labeling): Herpes simplex of the eye (except for short-term or emergency therapy); vaccinia and varicella (except for short-term or emergency therapy)

Warnings/Precautions

Concerns related to adverse effects:

• Adrenal suppression: May cause hypercortisolism or suppression of hypothalamic-pituitary-adrenal (HPA) axis, particularly in younger children or in patients receiving high doses for prolonged periods. HPA axis suppression may lead to adrenal crisis if corticosteroids are withdrawn. Withdrawal and discontinuation of a corticosteroid should be done slowly and carefully. Particular care is required when patients are transferred from systemic corticosteroids to inhaled products due to possible adrenal insufficiency or withdrawal from steroids, including an increase in allergic symptoms. Patients receiving >20 mg per day of prednisone (or equivalent) may be most susceptible. Fatalities have occurred due to adrenal insufficiency in asthmatic patients during and after transfer from systemic corticosteroids to aerosol steroids; aerosol steroids do **not** provide the systemic steroid needed to treat patients having trauma, surgery, or infections.

• Anaphylactoid reactions: Rare cases of anaphylactoid reactions have been observed in patients receiving corticosteroids.

• Dermal changes: Avoid injection or leakage into the dermis; dermal and/or subdermal skin depression may occur at the site of injection. Avoid deltoid muscle injection; subcutaneous atrophy may occur.

• Immunosuppression: Corticosteroids suppress the immune system and increase risk of infection with any pathogen, including bacterial, fungal, helminthic, protozoan, or viral; severe and sometimes fatal corticosteroid-associated infections have occurred. Corticosteroids may reduce resistance to new infections, exacerbate existing infections, increase the risk of disseminated infections, increase the risk of reactivation or exacerbation of latent infections, or mask some signs of infection. Avoid exposure to varicella or measles; if exposure occurs, prophylaxis with an appropriate immunoglobulin may be indicated; antiviral treatment may be considered if varicella infection occurs. Tuberculosis reactivation may occur when treating patients with latent tuberculosis or tuberculin reactivity; administer chemoprophylaxis in patients with latent tuberculosis or tuberculin reactivity during prolonged use of hydrocortisone. Avoid use in patients with active ocular herpes simplex. Hepatitis B reactivation can occur (infrequently) in patients who are hepatitis B carriers. For patients who show evidence of hepatitis B infection, consult an infectious disease specialist regarding monitoring and treatment options before initiating hydrocortisone. May exacerbate systemic fungal infections. Amebiasis reactivation may occur. Use with extreme caution in patients with *Strongyloides* infections; hyperinfection, dissemination, and fatalities have occurred. Avoid use in patients with cerebral malaria.

• Kaposi sarcoma: Prolonged treatment with corticosteroids has been associated with the development of Kaposi sarcoma; clinical improvement may occur with hydrocortisone discontinuation.

• Myopathy: Acute myopathy has been reported with high dose corticosteroids, usually in patients with neuromuscular transmission disorders; may involve ocular and/or respiratory muscles; monitor creatine kinase; recovery may be delayed.

• Psychiatric disturbances: Corticosteroid use may cause psychiatric disturbances, including euphoria, insomnia, mood swings, personality changes, severe depression, or psychotic manifestations. Symptoms usually occur within a few days or weeks of initiation of treatment and usually resolve with dose reduction or discontinuation; monitor for signs and symptoms of behavioral or mood changes. Preexisting psychiatric conditions may be exacerbated by corticosteroid use.

Disease-related concerns:

• Cardiovascular disease: Use with caution in patients with HF and/or hypertension; use has been associated with fluid retention, electrolyte disturbances, and hypertension. Use with caution following acute MI; corticosteroids have been associated with myocardial rupture.

• Diabetes: Use corticosteroids with caution in patients with diabetes mellitus; may alter glucose production/regulation leading to hyperglycemia.

• GI disease: Use with caution in patients with GI diseases (diverticulitis, fresh intestinal anastomoses, active or latent peptic ulcer, ulcerative colitis, abscess or other pyogenic infection) due to perforation risk; risk may be increased with concurrent use of nonsteroidal anti-inflammatory drugs.

• Head injury: Increased mortality was observed in patients receiving high-dose IV methylprednisolone; high-dose corticosteroids should not be used for the management of head injury.

• Hepatic impairment: Use with caution in patients with hepatic impairment, including cirrhosis; long-term use has been associated with fluid retention.

• Kidney impairment: Use with caution in patients with kidney impairment; fluid retention may occur.

• Myasthenia gravis: Use may cause transient worsening of myasthenia gravis (MG) (eg, within first 2 weeks of treatment); monitor for worsening MG (AAN [Narayanaswami 2021]).

• Ocular disease: Use with caution in patients with cataracts and/or glaucoma; increased intraocular pressure, open-angle glaucoma, and cataracts have occurred with prolonged use. Oral steroid treatment is not recommended for the treatment of acute optic neuritis; may increase frequency of new episodes and does not affect short- or long-term visual outcomes. Use with caution in patients with ocular herpes simplex; corneal perforation may occur; do not use in active ocular herpes simplex. Consider routine eye exams in chronic users.

• Osteoporosis: Use with caution in patients with osteoporosis; high doses and/or long-term use of corticosteroids have been associated with increased bone loss and osteoporotic fractures.

• Pheochromocytoma: Pheochromocytoma crisis has been reported with corticosteroids (may be fatal). Consider the risk of pheochromocytoma crisis prior to administering corticosteroids in patients with suspected pheochromocytoma.

• Seizure disorders: Use corticosteroids with caution in patients with a history of seizure disorder; seizures have been reported with adrenal crisis.

• Septic shock or sepsis syndrome: Corticosteroids should not be administered for the treatment of sepsis in the absence of shock (SCCM/ESICM [Annane 2017]).

• Systemic sclerosis: Use with caution in patients with systemic sclerosis; an increase in scleroderma kidney crisis incidence has been observed with corticosteroid use. Monitor BP and kidney function in patients with systemic sclerosis treated with corticosteroids (EULAR [Kowal-Bielecka 2017]).

• Thyroid disease: Changes in thyroid status may necessitate dosage adjustments; metabolic clearance of corticosteroids increases in patients with hyperthyroidism and decreases in patients with hypothyroidism.

Special populations:

• Older adults: Use with caution in the older adult with the smallest possible effective dose for the shortest duration.

• Pediatric: May affect growth velocity; growth should be routinely monitored in pediatric patients.

Dosage form specific issues:

• Benzyl alcohol and derivatives: Diluent for injection may contain benzyl alcohol and some dosage forms may contain sodium benzoate/benzoic acid; benzoic acid (benzoate) is a metabolite of benzyl alcohol; large amounts of benzyl alcohol (≥99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol and/or benzyl alcohol derivative with caution in neonates. See manufacturer's labeling.

• Capsule: Sprinkles are therapeutically equivalent to tablets; however, compounding tablets into a suspension, or crushing or splitting tablets could lead to a relative difference in exposure on the same nominal dose leading to symptoms of adrenal insufficiency. Dosage adjustments of oral sprinkles may be needed if adrenal insufficiency occurs.

• Propylene glycol: Some dosage forms may contain propylene glycol; large amounts are potentially toxic and have been associated with hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Zar 2007).

Other warnings/precautions:

• Discontinuation of therapy: Withdraw therapy with gradual tapering of dose.

• Epidural injection: Corticosteroids are not approved for epidural injection. Serious neurologic events (eg, spinal cord infarction, paraplegia, quadriplegia, cortical blindness, stroke), some resulting in death, have been reported with epidural injection of corticosteroids, with and without use of fluoroscopy.

• Immunizations: Avoid administration of live or live attenuated vaccines in patients receiving immunosuppressive doses of corticosteroids. Nonlive or inactivated vaccines may be administered, although the response cannot be predicted.

• Stress: Patients may require higher doses when subject to stress (ie, trauma, surgery, severe infection). Oral solution (Khindivi) is not approved for increased dosing during periods of stress or acute events; other hydrocortisone products should be used.

Warnings: Additional Pediatric Considerations

May cause osteoporosis (at any age) or inhibition of bone growth in pediatric patients. Use with caution in patients with osteoporosis. In a population-based study of children, risk of fracture was shown to be increased with >4 courses of corticosteroids; underlying clinical condition may also impact bone health and osteoporotic effect of corticosteroids (Leonard 2007). In premature neonates, reports of gastrointestinal perforation in the hydrocortisone treatment arm have resulted in the closure of two large bronchopulmonary dysplasia (BPD) clinical trials (Peltoniemi 2005; Watterberg 2004); concomitant use with indomethacin or ibuprofen may increase the risk and should be avoided in this population (Seri 2006). Increased intraocular pressure (IOP) may occur especially with prolonged use; in children, increased IOP has been shown to be dose dependent and produce a greater IOP in children <6 years than older children treated with ophthalmic dexamethasone (Lam 2005). Hypertrophic cardiomyopathy has been reported in premature neonates.

In premature neonates, the use of high-dose dexamethasone (approximately >0.5 mg/kg/day) for the prevention or treatment of BPD has been associated with adverse neurodevelopmental outcomes, including higher rates of cerebral palsy without additional clinical benefit over lower doses; current data does not support use of high doses. Data specific to hydrocortisone use in this population has shown that use <7 days of therapy does not appear to be associated with adverse neurodevelopmental outcomes (Needelman 2010). Early initiation (ie, PNA <24 hours) of low-dose hydrocortisone for BPD prevention was shown to improve survival without BPD in neonates born at GA <28 weeks and was not associated with a statistically significant difference in neurodevelopment at 2 years corrected age compared to placebo; a secondary analysis looking at neurodevelopment according to gestation showed a statistically significant improvement in neurodevelopmental outcomes in neonates at 24 to 25 weeks gestation receiving hydrocortisone; the incidence of moderate to severe neurodevelopmental impairment was 2% in this group compared to 18% in the placebo group; no differences in neurodevelopmental outcomes were seen in neonates at 26 to 27 weeks gestation (Baud 2016; Baud 2017; Baud 2019a). Similar to early initiation, when hydrocortisone was initiated later (ie, PNA 14 to 28 days) in neonates born at <30 weeks gestation, no adverse neurodevelopmental effects were seen at 2 years corrected age compared to placebo (Watterberg 2022).

Some dosage forms may contain propylene glycol; in neonates large amounts of propylene glycol delivered orally, intravenously (eg, >3,000 mg/day), or topically have been associated with potentially fatal toxicities which can include metabolic acidosis, seizures, renal failure, and CNS depression;

toxicities have also been reported in children and adults including hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Shehab 2009).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Capsule Sprinkle, Oral:

Alkindi Sprinkle: 0.5 mg, 1 mg, 2 mg, 5 mg

Solution, Oral:

Khindivi: 1 mg/mL (473 mL) [contains methylparaben (methyl p-hydroxybenzoate), polyethylene glycol (macrogol), propylene glycol, propylparaben (propyl p-hydroxybenzoate); berry flavor]

Solution Reconstituted, Injection, as sodium succinate [strength expressed as base, preservative free]:

Solu-CORTEF: 100 mg (1 ea); 250 mg (1 ea); 500 mg (1 ea); 1000 mg (1 ea)

Generic: 100 mg (1 ea)

Tablet, Oral, as base:

Cortef: 5 mg, 10 mg, 20 mg [scored]

Generic: 5 mg, 10 mg, 20 mg

Generic Equivalent Available: US

May be product dependent

Pricing: US

Capsule, sprinkles (Alkindi Sprinkle Oral)

0.5 mg (per each): $12.74

1 mg (per each): $25.48

2 mg (per each): $50.96

5 mg (per each): $127.40

Solution (Khindivi Oral)

1 mg/mL (per mL): $31.85

Solution (reconstituted) (Hydrocortisone Sod Suc (PF) Injection)

100 mg (per each): $18.84

Solution (reconstituted) (Solu-CORTEF Injection)

100 mg (per each): $29.77

250 mg (per each): $57.68

500 mg (per each): $120.67

1000 mg (per each): $241.26

Tablets (Cortef Oral)

5 mg (per each): $1.21

10 mg (per each): $2.04

20 mg (per each): $3.86

Tablets (Hydrocortisone Oral)

5 mg (per each): $0.22 - $1.02

10 mg (per each): $0.36 - $2.15

20 mg (per each): $0.62 - $3.26

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution Reconstituted, Injection, as sodium succinate [strength expressed as base]:

Solu-CORTEF: 100 mg (1 ea); 250 mg (1 ea); 500 mg (1 ea); 1000 mg (1 ea)

Tablet, Oral, as base:

Cortef: 10 mg, 20 mg [contains corn starch]

Generic: 10 mg, 20 mg

Extemporaneous Preparations

2 mg/mL Oral Suspension (ASHP Standard Concentration) (Ref)

A 2 mg/mL oral suspension may be made with 10 mg tablets and a 1:1 mixture of Ora-Sweet and Ora-Plus. Crush ten 10 mg hydrocortisone tablets in a mortar and reduce to a fine powder. Add a small amount of vehicle and mix to a uniform paste; mix while adding the vehicle in incremental proportions to **almost** 50 mL; transfer to a calibrated amber bottle, rinse mortar with vehicle, and add sufficient quantity of vehicle to make 50 mL. Label "shake well." Stable under refrigeration or at room temperature for 90 days (Ref).

2.5 mg/mL Oral Suspension

A 2.5 mg/mL oral suspension may be made with either tablets or powder and a vehicle containing sodium carboxymethylcellulose (1 g), syrup BP (10 mL), hydroxybenzoate 0.1% preservatives (0.1 g), polysorbate 80 (0.5 mL), citric acid (0.6 g), and water. To make the vehicle, dissolve the hydroxybenzoate, citric acid, and syrup BP in hot water. Cool solution and add the carboxymethylcellulose; leave overnight. Crush twelve-and-one-half 20 mg hydrocortisone tablets (or use 250 mg of powder) in a mortar and reduce to a fine powder while adding polysorbate 80. Add small portions of vehicle and mix to a uniform paste; mix while adding the vehicle in incremental proportions to **almost** 100 mL; transfer to a calibrated bottle, rinse mortar with vehicle, and add sufficient quantity of vehicle to make 100 mL. Label "shake well" and "refrigerate." Stable for 90 days (Ref),

Administration: Adult

Oral: Administer with food or milk to decrease GI upset.

Parenteral: For IM or IV administration: Dermal and/or subdermal skin depression may occur at injection site.

IM: Avoid injection into deltoid muscle (high incidence of subcutaneous atrophy).

IV bolus: Administer undiluted over at least 30 seconds; for large doses (≥500 mg), administer over 10 minutes.

IV intermittent infusion: Further dilute in a compatible fluid and administer over 20 to 30 minutes.

SUBQ (off-label route): Divide injections >1 mL (>50 mg) into 2 or more separate injections (Ref).

Preparation for Administration: Adult

IV bolus or IM administration: Reconstitute 100 mg vials with bacteriostatic water or bacteriostatic sodium chloride (not >2 mL).

IV infusion administration: Add reconstituted solutions to an appropriate volume of D5W, NS, or D5NS (100 to 1,000 mL for a 100 mg solution; 250 to 1,000 mL for a 250 mg solution; 500 to 1,000 mL for a 500 mg solution; 1,000 mL for a 1,000 mg solution). In cases where administration of a small volume of fluid is desirable, 100 to 3,000 mg of hydrocortisone may be added to 50 mL of D5W or NS.

SUBQ administration (off-label route): Reconstitute 100 mg vials with 2 mL bacteriostatic water or bacteriostatic sodium chloride (Ref).

Act-O-Vial: Press the activator to force diluent into the powder compartment. Following gentle agitation, solution may be withdrawn via syringe through a needle inserted into the center of the stopper.

Administration: Pediatric

Oral: Administer with food or milk to decrease GI upset; if vomiting occurs within an hour of the dose, repeat dose. For physiologic replacement in pediatric patients, higher doses are typically administered in the morning and midday with lower doses in the evening to replicate diurnal variation; early evening doses (18:00) may be necessary in some children (doses too close to bedtime can interfere with sleep) (Ref).

Oral solution (commercially available) (Khindivi): May be administered without regard to meals. Administer with an accurate measuring device (eg, calibrated oral syringe or measuring cup); do not use a household teaspoon or tablespoon to measure dose (overdosage may occur).

Administration via feeding tube:

Gastric (eg, NG, G-tube) tubes: Draw up hydrocortisone oral solution into enteral dosing syringe and administer via a feeding tube. Following administration, flush feeding tube with 20 mL of purified water to ensure the entire dose is delivered.

Sprinkles (Alkindi): Do not swallow capsule whole. Hold capsule so numeric strength section of the capsule is at the top and tap capsule to ensure all granules in lower section, then open the capsule by squeezing the bottom section and twisting the top off. Granules may be administered by any of the following methods: Pouring granules directly onto patient’s tongue; pouring granules onto a spoon and placing in patient’s mouth; or sprinkling granules onto a spoon containing a soft food (eg, yogurt, fruit puree) that is cold or at room temperature and then having patient consume preferably within 5 minutes to avoid bitter taste. Avoid getting capsule wet (if capsule is wet, granules may stick and not get delivered in dose). Do not crush or chew granules. Ingest fluids (eg, water, milk, breast milk, infant formula) after dose is administered to ensure all granules swallowed. Do not add granules directly to a liquid; may result in reduced dose delivered as well as a bitter taste.

***Administration via feeding tube: NOT* recommended.** Enteral feeding tube administration utilizing hydrocortisone sprinkles is not recommended; granules will clog tube.

Parenteral: Hydrocortisone sodium succinate may be administered by IM or IV routes. Dermal and/or subdermal skin depression may occur at the site of injection.

IM: Avoid injection into deltoid muscle (high incidence of SUBQ atrophy).

IV bolus: Administer undiluted over at least 30 seconds; for large doses (≥500 mg), administer over 10 minutes.

Intermittent IV infusion: Further dilute in a compatible fluid and administer over 20 to 30 minutes (Ref).

Preparation for Administration: Pediatric

Parenteral: Reconstitute vial with appropriate diluent, either bacteriostatic water or NS (see manufacturer's labeling for details). For the commonly used 100 mg vial, reconstitute with a volume of diluent not to exceed 2 mL resulting in a concentration ≥ 50 mg/mL. Act-O-Vial (self-contained powder for injection plus diluent [preservative free SWFI]) may be reconstituted by pressing the activator to force diluent into the powder compartment. Following gentle agitation, solution may be withdrawn via syringe through a needle inserted into the center of the stopper. May be administered (IV or IM) without further dilution.

Intermittent IV infusion: Reconstituted solutions may be further diluted in an appropriate volume of compatible solution for infusion. Concentration should generally not exceed 1 mg/mL. In pediatric patients, a concentration of 5 mg/mL has been used (Ref). The manufacturer suggests, in cases where administration of a small volume of fluid is desirable, concentrations up to 60 mg/mL (100 to 3,000 mg in 50 mL of D_5W or NS; stability limited to 4 hours) may be used.

Storage/Stability

Capsule: Store at 20°C to 25°C (68°F to 77°F); excursions permitted between 15°C and 30°C (59°F and 86°F). Store in original container to protect from light. Use within 60 days of opening bottle.

Injection: Store intact vials at 20°C to 25°C (68°F to 77°F). Heat sensitive; do not autoclave vial. Reconstituted vials and solutions diluted for infusion may be stored for up to 12 hours at 20°C to 25°C (68°F to 77°F) or up to 24 hours under refrigeration; protect from light. Discard any unused portion. (**Note:** Prior to August 2024, the prescribing information stated the reconstituted solution could be stored for up to 3 days at room temperature).

Oral solution: Store at 2°C to 25°C (36°F to 77°F); excursions permitted to 30°C (86°F). Protect from heat and light. Use within 120 days of initial opening; discard any unused portion.

Tablet: Store at 20°C to 25°C (68°F to 77°F).

Use: Labeled Indications

Allergic states: Control of severe or incapacitating allergic conditions intractable to adequate trials of conventional treatment in drug hypersensitivity reactions, perennial or seasonal allergic rhinitis (oral only), serum sickness, transfusion reactions, or acute noninfectious laryngeal edema (epinephrine is the drug of first choice).

Dermatologic diseases: Atopic dermatitis; bullous dermatitis herpetiformis; contact dermatitis; exfoliative dermatitis; exfoliative erythroderma; pemphigus; severe erythema multiforme (Stevens-Johnson syndrome); severe psoriasis; severe seborrheic dermatitis; mycosis fungoides.

Endocrine disorders: Acute adrenocortical insufficiency; congenital adrenal hyperplasia; hypercalcemia associated with cancer; nonsuppurative thyroiditis; primary or secondary adrenocortical insufficiency; preoperatively and in the event of serious trauma or illness, in patients with known adrenal insufficiency or when adrenocortical reserve is doubtful; shock unresponsive to conventional therapy if adrenocortical insufficiency exists or is suspected.

GI diseases: To tide the patient over a critical period of the disease in ulcerative colitis and regional enteritis.

Hematologic disorders: Acquired (autoimmune) hemolytic anemia; congenital (erythroid) hypoplastic anemia (Diamond Blackfan anemia); erythroblastopenia (RBC anemia); immune thrombocytopenia (formerly known as idiopathic thrombocytopenic purpura) in adults; pure red cell aplasia; select cases of secondary thrombocytopenia.

Neoplastic diseases: Palliative management of leukemias and lymphomas (adults); acute leukemia of childhood.

Ophthalmic diseases: Severe acute and chronic allergic and inflammatory processes involving the eye, such as allergic conjunctivitis; allergic corneal marginal ulcers; anterior segment inflammation; chorioretinitis; diffuse posterior uveitis and choroiditis; herpes zoster ophthalmicus; iritis and iridocyclitis; keratitis; optic neuritis; sympathetic ophthalmia; other ocular inflammatory conditions unresponsive to topical corticosteroids.

Respiratory diseases: Aspiration pneumonitis; bronchial asthma; berylliosis; fulminating or disseminated pulmonary tuberculosis when used concurrently with appropriate antituberculous chemotherapy; idiopathic eosinophilic pneumonias; Loeffler syndrome (not manageable by other means); symptomatic sarcoidosis.

Rheumatic disorders: As adjunctive therapy for short-term administration in acute and subacute bursitis, acute gouty arthritis, acute nonspecific tenosynovitis, ankylosing spondylitis, epicondylitis, posttraumatic osteoarthritis, psoriatic arthritis, rheumatoid arthritis, including juvenile rheumatoid arthritis, synovitis of osteoarthritis; during an exacerbation or as maintenance therapy in acute rheumatic carditis, dermatomyositis (polymyositis), temporal arteritis, and systemic lupus erythematosus.

Miscellaneous: Trichinosis with neurologic or myocardial involvement; tuberculous meningitis with subarachnoid block or impending block when used concurrently with appropriate antituberculous chemotherapy.

Use: Off-Label: Adult

COVID-19, hospitalized patients; Deceased organ donor management (hormonal replacement therapy); Pneumonia, community acquired, severe; Septic shock; Thyroid storm

Medication Safety Issues

Sound-alike/look-alike issues:

Hydrocortisone may be confused with hydrocodone, hydroxychloroquine, hydroCHLOROthiazide

Cortef may be confused with Coreg, Lortab

Solu-CORTEF may be confused with SOLU-Medrol

Other safety concerns:

HCT is an error-prone abbreviation (mistaken as hydroCHLOROthiazide)

Metabolism/Transport Effects

Substrate of CYP3A4 (Minor), P-glycoprotein (Minor); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Abrocitinib: Corticosteroids (Systemic) may increase immunosuppressive effects of Abrocitinib. Management: The use of abrocitinib in combination with other immunosuppressants is not recommended. Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks are considered immunosuppressive. *Risk D: Consider Therapy Modification*

Acetylcholinesterase Inhibitors: Corticosteroids (Systemic) may increase adverse/toxic effects of Acetylcholinesterase Inhibitors. Increased muscular weakness may occur. *Risk C: Monitor*

Aldesleukin: Corticosteroids (Systemic) may decrease therapeutic effects of Aldesleukin. *Risk X: Avoid*

Amphotericin B: Corticosteroids (Systemic) may increase hypokalemic effects of Amphotericin B. *Risk C: Monitor*

Androgens: Corticosteroids (Systemic) may increase fluid-retaining effects of Androgens. *Risk C: Monitor*

Antidiabetic Agents: Hyperglycemia-Associated Agents may decrease therapeutic effects of Antidiabetic Agents. *Risk C: Monitor*

Antithymocyte Globulin (Equine): Corticosteroids (Systemic) may increase adverse/toxic effects of Antithymocyte Globulin (Equine). Specifically, these effects may be unmasked if the dose of systemic corticosteroid is reduced. Corticosteroids (Systemic) may increase immunosuppressive effects of Antithymocyte Globulin (Equine). Specifically, infections may occur with greater severity and/or atypical presentations. *Risk C: Monitor*

Baricitinib: Corticosteroids (Systemic) may increase immunosuppressive effects of Baricitinib. Management: The use of baricitinib in combination with potent immunosuppressants is not recommended. Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks are considered immunosuppressive. *Risk D: Consider Therapy Modification*

BCG Products: Corticosteroids (Systemic) may decrease therapeutic effects of BCG Products. Corticosteroids (Systemic) may increase adverse/toxic effects of BCG Products. Specifically, the risk of vaccine-associated infection may be increased. *Risk X: Avoid*

Bile Acid Sequestrants: May decrease absorption of Hydrocortisone (Systemic). *Risk C: Monitor*

Brincidofovir: Corticosteroids (Systemic) may decrease therapeutic effects of Brincidofovir. *Risk C: Monitor*

Brivudine: May increase adverse/toxic effects of Corticosteroids (Systemic). *Risk X: Avoid*

Calcitriol (Systemic): Corticosteroids (Systemic) may decrease therapeutic effects of Calcitriol (Systemic). *Risk C: Monitor*

CAR-T Cell Immunotherapy: Corticosteroids (Systemic) may decrease therapeutic effects of CAR-T Cell Immunotherapy. Corticosteroids (Systemic) may increase adverse/toxic effects of CAR-T Cell Immunotherapy. Specifically, the severity and duration of neurologic toxicities may be increased. Management: Avoid use of corticosteroids as premedication before treatment with CAR-T cell immunotherapy agents. Corticosteroids are indicated and may be required for treatment of toxicities such as cytokine release syndrome or neurologic toxicity. *Risk D: Consider Therapy Modification*

Cardiac Glycosides: Corticosteroids (Systemic) may increase adverse/toxic effects of Cardiac Glycosides. *Risk C: Monitor*

Chikungunya Vaccine (Live): Corticosteroids (Systemic) may increase adverse/toxic effects of Chikungunya Vaccine (Live). Specifically, the risk of vaccine-associated infection may be increased. Corticosteroids (Systemic) may decrease therapeutic effects of Chikungunya Vaccine (Live). *Risk X: Avoid*

Cladribine: Corticosteroids (Systemic) may increase immunosuppressive effects of Cladribine. *Risk X: Avoid*

Coccidioides immitis Skin Test: Coadministration of Corticosteroids (Systemic) and Coccidioides immitis Skin Test may alter diagnostic results. Management: Consider discontinuing systemic corticosteroids (dosed at 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks) several weeks prior to coccidioides immitis skin antigen testing. *Risk D: Consider Therapy Modification*

Cosyntropin: Coadministration of Hydrocortisone (Systemic) and Cosyntropin may alter diagnostic results. Management: Patients receiving hydrocortisone should omit their pre-test dose on the day selected for cosyntropin testing. *Risk D: Consider Therapy Modification*

COVID-19 Vaccine (Inactivated Virus): Corticosteroids (Systemic) may decrease therapeutic effects of COVID-19 Vaccine (Inactivated Virus). *Risk C: Monitor*

COVID-19 Vaccine (mRNA): Corticosteroids (Systemic) may decrease therapeutic effects of COVID-19 Vaccine (mRNA). Management: Give a 3-dose primary series for all patients aged 6 months and older taking immunosuppressive medications or therapies. Booster doses are recommended for certain age groups. See CDC guidance for details. *Risk D: Consider Therapy Modification*

COVID-19 Vaccine (Subunit): Corticosteroids (Systemic) may decrease therapeutic effects of COVID-19 Vaccine (Subunit). *Risk C: Monitor*

CYP3A4 Inducers (Moderate): May decrease serum concentrations of Hydrocortisone (Systemic). *Risk C: Monitor*

CYP3A4 Inducers (Strong): May decrease serum concentrations of Hydrocortisone (Systemic). *Risk C: Monitor*

CYP3A4 Inducers (Weak): May decrease serum concentrations of Hydrocortisone (Systemic). *Risk C: Monitor*

CYP3A4 Inhibitors (Moderate): May increase serum concentrations of Hydrocortisone (Systemic). *Risk C: Monitor*

CYP3A4 Inhibitors (Strong): May increase serum concentrations of Hydrocortisone (Systemic). *Risk C: Monitor*

Deferasirox: Corticosteroids (Systemic) may increase adverse/toxic effects of Deferasirox. Specifically, the risk for GI ulceration/irritation or GI bleeding may be increased. *Risk C: Monitor*

Delgocitinib: May increase immunosuppressive effects of Corticosteroids (Systemic). Management: The use of delgocitinib in combination with other immunosuppressants is not recommended. Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks are considered immunosuppressive. *Risk D: Consider Therapy Modification*

Dengue Tetravalent Vaccine (Live): Corticosteroids (Systemic) may decrease therapeutic effects of Dengue Tetravalent Vaccine (Live). Corticosteroids (Systemic) may increase adverse/toxic effects of Dengue Tetravalent Vaccine (Live). Specifically, the risk of vaccine associated infection may be increased. *Risk X: Avoid*

Denosumab: May increase immunosuppressive effects of Corticosteroids (Systemic). Management: Consider the risk of serious infections versus the potential benefits of coadministration of denosumab and systemic corticosteroids. If combined, monitor patients for signs/symptoms of serious infections. *Risk D: Consider Therapy Modification*

Desmopressin: Corticosteroids (Systemic) may increase hyponatremic effects of Desmopressin. *Risk X: Avoid*

Deucravacitinib: May increase immunosuppressive effects of Corticosteroids (Systemic). Management: The use of deucravacitinib in combination with potent immunosuppressants is not recommended. Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks are considered immunosuppressive. *Risk D: Consider Therapy Modification*

Dinutuximab Beta: Corticosteroids (Systemic) may increase immunosuppressive effects of Dinutuximab Beta. Management: Corticosteroids are not recommended for 2 weeks prior to dinutuximab beta, during therapy and for 1 week after treatment. Doses equivalent to over 2 mg/kg or 20 mg/day of prednisone (persons over 10 kg) for 2 or more weeks are considered immunosuppressive *Risk D: Consider Therapy Modification*

Estrogen Derivatives: May increase serum concentrations of Hydrocortisone (Systemic). Estrogen Derivatives may decrease serum concentrations of Hydrocortisone (Systemic). *Risk C: Monitor*

Etrasimod: Corticosteroids (Systemic) may increase immunosuppressive effects of Etrasimod. *Risk C: Monitor*

Filgotinib: Corticosteroids (Systemic) may increase immunosuppressive effects of Filgotinib. Management: Coadministration of filgotinib with systemic corticosteroids at doses equivalent to greater than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks is not recommended. *Risk D: Consider Therapy Modification*

Gallium Ga 68 Dotatate: Coadministration of Corticosteroids (Systemic) and Gallium Ga 68 Dotatate may alter diagnostic results. *Risk C: Monitor*

Grapefruit Juice: May increase serum concentrations of Hydrocortisone (Systemic). *Risk C: Monitor*

Growth Hormone Analogs: Corticosteroids (Systemic) may decrease therapeutic effects of Growth Hormone Analogs. Growth Hormone Analogs may decrease active metabolite exposure of Corticosteroids (Systemic). *Risk C: Monitor*

Histamine: And Corticosteroids (Systemic) may interact via an unclear mechanism. *Risk C: Monitor*

Hyaluronidase: Corticosteroids (Systemic) may decrease therapeutic effects of Hyaluronidase. Management: Patients receiving corticosteroids (particularly at larger doses) may not experience the desired clinical response to standard doses of hyaluronidase. Larger doses of hyaluronidase may be required. *Risk D: Consider Therapy Modification*

Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies): Corticosteroids (Systemic) may decrease therapeutic effects of Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies). Management: Carefully consider the need for corticosteroids, at doses of a prednisone-equivalent of 10 mg or more per day, during the initiation of immune checkpoint inhibitor therapy. Use of corticosteroids to treat immune related adverse events is still recommended *Risk D: Consider Therapy Modification*

Indium 111 Capromab Pendetide: Coadministration of Corticosteroids (Systemic) and Indium 111 Capromab Pendetide may alter diagnostic results. Management: Consider alternatives when possible, as product labeling states the concomitant use of these agents is not recommended. If combined, consider that the diagnostic results obtained with indium 111 capromab pendetide may be altered or unreliable. *Risk D: Consider Therapy Modification*

Inebilizumab: Corticosteroids (Systemic) may increase immunosuppressive effects of Inebilizumab. *Risk C: Monitor*

Influenza Virus Vaccines: Corticosteroids (Systemic) may decrease therapeutic effects of Influenza Virus Vaccines. Management: Administer influenza vaccines at least 2 weeks prior to initiation of systemic corticosteroids at immunosuppressive doses. Influenza vaccines administered less than 14 days prior to or during such therapy should be repeated 3 months after therapy. *Risk D: Consider Therapy Modification*

Isoniazid: Corticosteroids (Systemic) may decrease serum concentrations of Isoniazid. *Risk C: Monitor*

Leflunomide: Corticosteroids (Systemic) may increase immunosuppressive effects of Leflunomide. Management: Increase the frequency of chronic monitoring of platelet, white blood cell count, and hemoglobin or hematocrit to monthly, instead of every 6 to 8 weeks, if leflunomide is coadministered with immunosuppressive agents, such as systemic corticosteroids. *Risk D: Consider Therapy Modification*

Licorice: May increase serum concentrations of Corticosteroids (Systemic). *Risk C: Monitor*

Loop Diuretics: Corticosteroids (Systemic) may increase hypokalemic effects of Loop Diuretics. *Risk C: Monitor*

Lutetium Lu 177 Dotatate: Corticosteroids (Systemic) may decrease therapeutic effects of Lutetium Lu 177 Dotatate. Management: Avoid repeated use of high-doses of corticosteroids during treatment with lutetium Lu 177 dotatate. Use of corticosteroids is still permitted for the treatment of neuroendocrine hormonal crisis. The effects of lower corticosteroid doses is unknown. *Risk D: Consider Therapy Modification*

Macimorelin: Coadministration of Corticosteroids (Systemic) and Macimorelin may alter diagnostic results. *Risk X: Avoid*

MetyraPONE: Coadministration of Corticosteroids (Systemic) and MetyraPONE may alter diagnostic results. Management: Consider alternatives to the use of the metyrapone test in patients taking systemic corticosteroids. *Risk D: Consider Therapy Modification*

Mifamurtide: Corticosteroids (Systemic) may decrease therapeutic effects of Mifamurtide. *Risk X: Avoid*

MiFEPRIStone: May decrease therapeutic effects of Corticosteroids (Systemic). MiFEPRIStone may increase serum concentrations of Corticosteroids (Systemic). Management: Avoid mifepristone in patients who require long-term corticosteroid treatment of serious illnesses or conditions (eg, for immunosuppression following transplantation). Corticosteroid effects may be reduced by mifepristone treatment. *Risk X: Avoid*

Mumps- Rubella- or Varicella-Containing Live Vaccines: Corticosteroids (Systemic) may decrease therapeutic effects of Mumps- Rubella- or Varicella-Containing Live Vaccines. Corticosteroids (Systemic) may increase adverse/toxic effects of Mumps- Rubella- or Varicella-Containing Live Vaccines. Specifically, the risk of vaccine-associated infection may be increased. *Risk X: Avoid*

Nadofaragene Firadenovec: Corticosteroids (Systemic) may increase adverse/toxic effects of Nadofaragene Firadenovec. Specifically, the risk of disseminated adenovirus infection may be increased. *Risk X: Avoid*

Natalizumab: Corticosteroids (Systemic) may increase immunosuppressive effects of Natalizumab. *Risk X: Avoid*

Neuromuscular-Blocking Agents (Nondepolarizing): May increase adverse neuromuscular effects of Corticosteroids (Systemic). Increased muscle weakness, possibly progressing to polyneuropathies and myopathies, may occur. Management: If concomitant therapy is required, use the lowest dose for the shortest duration to limit the risk of myopathy or neuropathy. Monitor for new onset or worsening muscle weakness, reduction or loss of deep tendon reflexes, and peripheral sensory decriments *Risk D: Consider Therapy Modification*

Nicorandil: Corticosteroids (Systemic) may increase adverse/toxic effects of Nicorandil. Gastrointestinal perforation has been reported in association with this combination. *Risk C: Monitor*

Nirmatrelvir and Ritonavir: May increase serum concentrations of Hydrocortisone (Systemic). *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (COX-2 Selective): Corticosteroids (Systemic) may increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents (COX-2 Selective). *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (Nonselective): Corticosteroids (Systemic) may increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents (Nonselective). *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (Topical): May increase adverse/toxic effects of Corticosteroids (Systemic). Specifically, the risk of gastrointestinal bleeding, ulceration, and perforation may be increased. *Risk C: Monitor*

Ocrelizumab: Corticosteroids (Systemic) may increase immunosuppressive effects of Ocrelizumab. *Risk C: Monitor*

Ofatumumab: Corticosteroids (Systemic) may increase immunosuppressive effects of Ofatumumab. *Risk C: Monitor*

Ozanimod: Corticosteroids (Systemic) may increase immunosuppressive effects of Ozanimod. *Risk C: Monitor*

Pidotimod: Corticosteroids (Systemic) may decrease therapeutic effects of Pidotimod. *Risk C: Monitor*

Pimecrolimus: May increase immunosuppressive effects of Corticosteroids (Systemic). *Risk X: Avoid*

Pneumococcal Vaccines: Corticosteroids (Systemic) may decrease therapeutic effects of Pneumococcal Vaccines. *Risk C: Monitor*

Poliovirus Vaccines (Live/Oral): Corticosteroids (Systemic) may decrease therapeutic effects of Poliovirus Vaccines (Live/Oral). Corticosteroids (Systemic) may increase adverse/toxic effects of Poliovirus Vaccines (Live/Oral). Specifically, the risk of vaccine-associated infection may be increased. *Risk X: Avoid*

Polymethylmethacrylate: Corticosteroids (Systemic) may increase adverse/toxic effects of Polymethylmethacrylate. Specifically, the risk for hypersensitivity or implant clearance may be increased. Management: Use caution when considering use of bovine collagen-containing implants such as the polymethylmethacrylate-based Bellafill brand implant in patients who are receiving immunosuppressants. Consider use of additional skin tests prior to administration. *Risk D: Consider Therapy Modification*

Quinolones: Corticosteroids (Systemic) may increase adverse/toxic effects of Quinolones. Specifically, the risk of tendonitis and tendon rupture may be increased. *Risk C: Monitor*

Rabies Vaccine: Corticosteroids (Systemic) may decrease therapeutic effects of Rabies Vaccine. Management: Complete rabies vaccination at least 2 weeks before initiation of immunosuppressant therapy if possible. If combined, check for rabies antibody titers, and if vaccination is for post exposure prophylaxis, administer a 5th dose of the vaccine. *Risk D: Consider Therapy Modification*

Relacorilant: May decrease therapeutic effects of Corticosteroids (Systemic). Relacorilant may increase serum concentrations of Corticosteroids (Systemic). Management: Using relacorilant with corticosteroids used for lifesaving purposes is contraindicated. Avoid using recorilant with corticosteroids used for other conditions if possible. If combined, monitor for decreased corticosteroid and relacorilant effectiveness. *Risk D: Consider Therapy Modification*

Ritlecitinib: Corticosteroids (Systemic) may increase immunosuppressive effects of Ritlecitinib. Management: Use of ritlecitinib and immunosuppressants is not recommended. Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks are considered immunosuppressive. *Risk D: Consider Therapy Modification*

Ritodrine: Corticosteroids (Systemic) may increase adverse/toxic effects of Ritodrine. *Risk C: Monitor*

Ruxolitinib (Topical): Corticosteroids (Systemic) may increase immunosuppressive effects of Ruxolitinib (Topical). Management: The use of topical ruxolitinib and potent immunosuppressants is not recommended. Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks are considered immunosuppressive. *Risk D: Consider Therapy Modification*

Salicylates: May increase adverse/toxic effects of Corticosteroids (Systemic). These specifically include gastrointestinal ulceration and bleeding. Corticosteroids (Systemic) may decrease serum concentrations of Salicylates. Withdrawal of corticosteroids may result in salicylate toxicity. *Risk C: Monitor*

Sargramostim: Corticosteroids (Systemic) may increase therapeutic effects of Sargramostim. Specifically, corticosteroids may enhance the myeloproliferative effects of sargramostim. *Risk C: Monitor*

Sibeprenlimab: May increase immunosuppressive effects of Corticosteroids (Systemic). *Risk C: Monitor*

Sipuleucel-T: Corticosteroids (Systemic) may decrease therapeutic effects of Sipuleucel-T. Management: Consider reducing the dose or discontinuing immunosuppressants, such as systemic corticosteroids, prior to initiating sipuleucel-T therapy. Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone given for 2 or more weeks are immunosuppressive. *Risk D: Consider Therapy Modification*

Sitafloxacin: Corticosteroids (Systemic) may increase adverse/toxic effects of Sitafloxacin. Specifically, the risk of tendonitis and tendon rupture may be increased. Management: Consider alternatives to coadministration of corticosteroids and sitafloxacin unless the benefits of coadministration outweigh the risk of tendon disorders. *Risk D: Consider Therapy Modification*

Sodium Benzoate: Corticosteroids (Systemic) may decrease therapeutic effects of Sodium Benzoate. *Risk C: Monitor*

Sphingosine 1-Phosphate (S1P) Receptor Modulators: May increase immunosuppressive effects of Corticosteroids (Systemic). *Risk C: Monitor*

Succinylcholine: Corticosteroids (Systemic) may increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Tacrolimus (Systemic): Corticosteroids (Systemic) may decrease serum concentrations of Tacrolimus (Systemic). Conversely, when discontinuing corticosteroid therapy, tacrolimus concentrations may increase. *Risk C: Monitor*

Tacrolimus (Topical): Corticosteroids (Systemic) may increase immunosuppressive effects of Tacrolimus (Topical). *Risk X: Avoid*

Talimogene Laherparepvec: Corticosteroids (Systemic) may increase adverse/toxic effects of Talimogene Laherparepvec. Specifically, the risk of infection from the live, attenuated herpes simplex virus contained in talimogene laherparepvec may be increased. *Risk X: Avoid*

Tertomotide: Corticosteroids (Systemic) may decrease therapeutic effects of Tertomotide. *Risk X: Avoid*

Thiazide and Thiazide-Like Diuretics: Corticosteroids (Systemic) may increase hypokalemic effects of Thiazide and Thiazide-Like Diuretics. *Risk C: Monitor*

Tofacitinib: Corticosteroids (Systemic) may increase immunosuppressive effects of Tofacitinib. Management: Coadministration of tofacitinib with potent immunosuppressants is not recommended. Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks are considered immunosuppressive. *Risk D: Consider Therapy Modification*

Typhoid Vaccine: Corticosteroids (Systemic) may decrease therapeutic effects of Typhoid Vaccine. Corticosteroids (Systemic) may increase adverse/toxic effects of Typhoid Vaccine. Specifically, the risk of vaccine-associated infection may be increased. *Risk X: Avoid*

Ublituximab: Corticosteroids (Systemic) may increase immunosuppressive effects of Ublituximab. *Risk C: Monitor*

Upadacitinib: Corticosteroids (Systemic) may increase immunosuppressive effects of Upadacitinib. Management: Coadministration of upadacitinib with systemic corticosteroids at doses equivalent to greater than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks is not recommended. *Risk D: Consider Therapy Modification*

Urea Cycle Disorder Agents: Corticosteroids (Systemic) may decrease therapeutic effects of Urea Cycle Disorder Agents. More specifically, Corticosteroids (Systemic) may increase protein catabolism and plasma ammonia concentrations, thereby increasing the doses of Urea Cycle Disorder Agents needed to maintain these concentrations in the target range. *Risk C: Monitor*

Vaccines (Live): Corticosteroids (Systemic) may increase adverse/toxic effects of Vaccines (Live). Specifically, the risk of vaccine-associated infection may be increased. Corticosteroids (Systemic) may decrease therapeutic effects of Vaccines (Live). Management: Avoid live vaccines during and for 1 month after therapy with immunosuppressive doses of corticosteroids (equivalent to prednisone > 2 mg/kg or 20 mg/day in persons over 10 kg for at least 2 weeks). Give live vaccines 4 weeks prior to therapy if possible. *Risk D: Consider Therapy Modification*

Vaccines (Non-Live/Inactivated/Non-Replicating): Corticosteroids (Systemic) may decrease therapeutic effects of Vaccines (Non-Live/Inactivated/Non-Replicating). Management: Administer vaccines at least 2 weeks prior to immunosuppressive corticosteroids if possible. If patients are vaccinated less than 14 days prior to or during such therapy, repeat vaccination at least 3 months after therapy if immunocompetence restored. *Risk D: Consider Therapy Modification*

Vitamin K Antagonists: Corticosteroids (Systemic) may increase anticoagulant effects of Vitamin K Antagonists. *Risk C: Monitor*

Yellow Fever Vaccine: Corticosteroids (Systemic) may increase adverse/toxic effects of Yellow Fever Vaccine. Specifically, the risk of vaccine-associated infection may be increased. Corticosteroids (Systemic) may decrease therapeutic effects of Yellow Fever Vaccine. *Risk X: Avoid*

Zoster Vaccine (Live/Attenuated): Corticosteroids (Systemic) may increase adverse/toxic effects of Zoster Vaccine (Live/Attenuated). Specifically, the risk of vaccine-associated infection may be increased. Corticosteroids (Systemic) may decrease therapeutic effects of Zoster Vaccine (Live/Attenuated). *Risk X: Avoid*

Pregnancy Considerations

Some studies have shown an association between first trimester systemic corticosteroid use and oral clefts or decreased birth weight; however, information is conflicting and may be influenced by maternal dose/indication for use (Lunghi 2010; Park-Wyllie 2000; Pradat 2003). Hypoadrenalism may occur in newborns following maternal use of corticosteroids in pregnancy; monitor.

When treating patients with adrenal insufficiency (primary or central or congenital adrenal hyperplasia) during pregnancy, hydrocortisone is the preferred corticosteroid. Doses may need to be adjusted as pregnancy progresses. Pregnant patients with adrenal insufficiency should be monitored at least once each trimester (ES [Bornstein 2016]; ES [Fleseriu 2016]; ES [Speiser 2018]).

High doses of hydrocortisone (ie, stress doses) may be required to prevent adrenal crisis during labor in patients with known or suspected adrenal insufficiency (eg, Cushingoid appearance, prolonged glucocorticoid therapy) (ES [Bornstein 2016]; ES [Fleseriu 2016]; ES [Speiser 2018]; ESE [Luger 2021]). Patients who require systemic corticosteroids for management of asthma should also be given IV corticosteroids, such as hydrocortisone, during labor and for 24 hours after delivery to prevent adrenal crisis (ACOG [Dombrowski 2008]; ERS/TSANZ [Middleton 2020]). Patients using chronic low dose steroids for rheumatic disorders generally do not need a stress dose of hydrocortisone at the time of vaginal delivery; however, stress dosing may be needed in women with multiple comorbidities treated with chronic high dose steroid therapy. Stress dosing is recommended prior to cesarean delivery (ACR [Sammaritano 2020]).

Uncontrolled asthma is associated with adverse events on pregnancy (increased risk of perinatal mortality, preeclampsia, preterm birth, low birth weight infants, cesarean delivery, and the development of gestational diabetes). Poorly controlled asthma or asthma exacerbations may have a greater fetal/maternal risk than what is associated with appropriately used asthma medications. Maternal treatment improves pregnancy outcomes by reducing the risk of some adverse events (eg, preterm birth, gestational diabetes). Maternal asthma symptoms should be monitored monthly during pregnancy. Inhaled corticosteroids are recommended for the treatment of asthma during pregnancy; however, systemic corticosteroids should be used to control acute exacerbations or treat severe persistent asthma. Hydrocortisone may be used when parenteral administration is required (ERS/TSANZ [Middleton 2020]; GINA 2024).

Hydrocortisone is approved for the treatment of rheumatic disorders, however, when systemic corticosteroids are needed in pregnancy, nonfluorinated corticosteroids (eg, prednisone) are preferred (ACR [Sammaritano 2020]).

For dermatologic disorders in pregnant patients, systemic corticosteroids are generally not preferred for initial therapy; should be avoided during the first trimester; and used during the second or third trimester at the lowest effective dose (Leachman 2006).

In nonpregnant patients, hydrocortisone is recommended off label as an alternative corticosteroid for the management of COVID-19 (NIH 2023). In general, the treatment of COVID-19 during pregnancy is the same as in nonpregnant patients. However, because data for most therapeutic agents in pregnant patients are limited, treatment options should be evaluated as part of a shared decision-making process. Hydrocortisone is also an alternative for use in pregnant patients with severe or critical COVID-19 due to limited placental transfer. Suggested treatment algorithms are available for pregnant patients with severe or critical COVID-19 requiring corticosteroids (Saad 2020; Teelucksingh 2022). The risk of severe illness from COVID-19 infection is increased in symptomatic pregnant patients compared to nonpregnant patients (ACOG 2023). Information related to the treatment of COVID-19 during pregnancy continues to emerge; refer to current guidelines for the treatment of pregnant patients.

Breastfeeding Considerations

Corticosteroids are present in breast milk.

The manufacturer notes that when used systemically, maternal use of corticosteroids have the potential to cause adverse events in a breastfed infant (eg, growth suppression, interfere with endogenous corticosteroid production).

Single doses of hydrocortisone are considered acceptable for use in breastfeeding patients; data is not available following prolonged use. Corticosteroids are generally considered acceptable in breastfeeding patients when used in usual doses; however, monitoring of the infant is recommended (WHO 2002).

If there is concern about exposure to the infant, some guidelines recommend waiting 4 hours after the maternal dose of an oral systemic corticosteroid before breastfeeding in order to decrease potential exposure to the breastfed infant (based on a study using prednisolone) (ACR [Sammaritano 2020]; Butler 2014; ERS/TSANZ [Middleton 2020]; Leachman 2006; Ost 1985).

Dietary Considerations

Systemic use of corticosteroids may require a diet with increased potassium, vitamins A, B_6, C, D, folate, calcium, zinc, phosphorus, and decreased sodium. Some products may contain sodium.

Monitoring Parameters

Serum glucose, electrolytes; BP, weight, presence of infection; monitor IOP with therapy >6 weeks; bone mineral density; assess HPA axis suppression (eg, ACTH stimulation test, morning plasma cortisol test, urinary free cortisol test); signs and symptoms of behavioral or mood changes; signs and symptoms of Cushing syndrome every 6 months (pediatric patients <1 year of age may require monitoring every 3 to 4 months); growth in pediatric patients; screen for hepatitis B prior to initiation; latent or active amebiasis prior to initiation in patients who have spent time in tropical climates or have unexplained diarrhea; reactivation of tuberculosis in patients with latent tuberculosis or tuberculin reactivity.

Mechanism of Action

Short-acting corticosteroid with minimal sodium-retaining potential; decreases inflammation by suppression of migration of polymorphonuclear leukocytes and reversal of increased capillary permeability

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: IV: 1 hour.

Absorption: Rapid.

Bioavailability: Oral: 96% ± 20% (Czock 2005); oral sprinkles: ~87%.

Distribution: V_d: IV: 27 ± 7 L (Czock 2005); oral solution: Mean: 23 L (range: 5 to 59 L).

Protein binding: IV: 92% ± 2% (Czock 2005); oral solution/sprinkles: ≥90%.

Metabolism: Hepatic.

Half-life elimination:

IM: 2.2 ± 1.5 hours (Hahner 2013).

IV: 2 ± 0.3 hours.

Oral: 1.8 ± 0.5 hours (Czock 2005).

Oral solution: Mean: 0.95 hours (range: 0.3 to 2.3 hours).

SUBQ: 4.7 ± 4.7 hours (Hahner 2013).

Time to peak, plasma:

IM: 66 ± 51 minutes (Hahner 2013).

Oral: 1.2 ± 0.4 hours (Czock 2005).

Oral solution: Mean: 0.5 hours (range: 0.25 to 0.75 hours).

SUBQ: 91 ± 34 minutes (Hahner 2013).

Excretion: Urine (Czock 2005).

Medication Guide and/or Vaccine Information Statement (VIS)

An FDA-approved patient medication guide, which is available with the product information and as follows, must be dispensed with this medication:

Alkindi Sprinkle: https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/213876s001lbl.pdf#page=14

Khindivi: https://www.accessdata.fda.gov/drugsatfda_docs/label/2025/218980s000lbl.pdf

Patient Counseling Points

(For additional information see ”Hydrocortisone (systemic): Patient drug information”)

What is this drug used for?

• It is used for many health problems like allergy signs, asthma, adrenal gland problems, blood problems, skin rashes, or swelling problems. This is not a list of all health problems that this drug may be used for. Talk with the doctor.

Capsules and oral solution:

• This form of this drug is not approved for use in adults. However, your doctor may decide the benefits of taking this drug outweigh the risks. If you have been given this form of this drug, ask your doctor for information about the benefits and risks. Talk with your doctor if you have questions about taking this drug.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Upset stomach or throwing up

• Trouble sleeping

• Restlessness

• Sweating a lot

• Increased appetite

• Weight gain

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Infection like fever, chills, very bad sore throat, ear or sinus pain, cough, more sputum or change in color of sputum, pain with passing urine, mouth sores, or wound that will not heal

• High blood sugar like confusion, feeling sleepy, unusual thirst or hunger, passing urine more often, flushing, fast breathing, or breath that smells like fruit

• Low potassium levels like muscle pain or weakness, muscle cramps, or a heartbeat that does not feel normal

• Pancreas problem (pancreatitis) like very bad stomach pain, very bad back pain, or very bad upset stomach or throwing up

• High blood pressure like very bad headache or dizziness, passing out, or change in eyesight

• Cushing’s syndrome like weight gain in the upper back or belly, moon face, severe headache, or slow healing

• Weak adrenal gland like severe upset stomach or throwing up, severe dizziness or passing out, muscle weakness, feeling very tired, mood changes, decreased appetite, or weight loss

• Shortness of breath, a big weight gain, or swelling in the arms or legs

• Skin changes (pimples, stretch marks, slow healing, hair growth)

• Purple, red, blue, brown, or black bumps or patches on the skin or in the mouth

• Bone or joint pain

• Period (menstrual) changes

• Chest pain or pressure

• Change in eyesight, eye pain, or severe eye irritation

• Change in the way you act

• Depression or other mood changes

• Hallucinations (seeing or hearing things that are not there)

• Seizures

• Any unexplained bruising or bleeding

• Severe stomach pain

• Black, tarry, or bloody stools

• Throwing up blood or throw up that looks like coffee grounds

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Cortef | Flazole | Flebocortid | Hyԁrocortisone;
- (AR) Argentina: Fridalit | Hidrocort | Hidrocortisona | Hidrocortisona celtyc | Hidrocortisona drawer | Hidrocortisona fabra | Hidrocortisona Klonal | Hidrocortisona pharmavial | Hidrocortisona richet | Hidrotisona | Oralsone;
- (AT) Austria: Alkindi | Efmody | Hydrocortison | Hydrocortone;
- (AU) Australia: Hyԁrocortisone mylan | Hyԁrocortisone Sodium | Hyԁrocortisone viatris | Hydrocortsone | Hysone | Solu cortef;
- (BD) Bangladesh: Cortaid | Cortef | Cortinex | Cotson | Eucot | Glucort | Hydrason | Hyԁrocortisone | Inflanil | Rapicort;
- (BE) Belgium: Cortril | Hyԁrocortisone | Solu cortef;
- (BF) Burkina Faso: Hyԁrocortisone biocodex;
- (BG) Bulgaria: Alkindi | Hydrocortison | Hydrocortone;

- (BR) Brazil: Androcortil | Ariscorten | Benzenil | Cortisonal | Cortiston | Cortizol | Cortizon | Flebocortid | Gliocort | Hidrocortex | Hidrocortisona | Hidrosone | Solu cortef | Succinato sodico de hidrocortisona;
- (CH) Switzerland: Alkindi | Hydrocortison galepharm | Hydrocortone | Plenadren | Solu cortef;
- (CI) Côte d'Ivoire: Hycort | Hyԁrocortisone biocodex;
- (CL) Chile: Cortisol | Hialub | Hidrocortisona;
- (CN) China: Hydrocor.sod.succi | Hyԁrocortisone;
- (CO) Colombia: Hidrocortisona | Itrocsona | Kortafgor | Limicort | Solu cortef | Sumicort | Vesacort;
- (CR) Costa Rica: Hidrocortisona | Hidrocortisona succinato sodico | Lycortin-s | Vesacort;
- (CZ) Czech Republic: Cortef | Hydrocortison | Hydrocortison Vuab | Solucortef;
- (DE) Germany: Acecort | Alkindi | Efmody | Ficortril | Hydrocortipuren | Hydrocortison | Hydrocortison acis | Hydrocortison Galen | Hydrocortison hoechst | Hydrocortison Hoechst | Hydrocutan | Hydroson | Plenadren;
- (DO) Dominican Republic: Hidrocortisona | Humera;
- (EC) Ecuador: Hidrocortif | Hidrocortisona | Hidrocortisona klonal | Hidrocortisona succinato de sodio | Hidrocortisona succinato sodico | Solucortef | Succinato sodico de hidrocortisona;
- (EE) Estonia: Abix | Ampikyy | Hydrocortison | Solucortef;
- (EG) Egypt: Hyԁrocortisone | Micort | Sigmacortin | Solu cortef | Solucortef;
- (ES) Spain: Actocortina | Hidroaltesona | Hidrocortisona lorien | Hidrocortisona pharmis;
- (ET) Ethiopia: Cort S;
- (FI) Finland: Ampikyy | Efmody | Hydro adreson | Hydrocortison | Hyԁrocortisone panpharma | Kyypakkaus | Plenadren | Solu cortef;
- (FR) France: Hyԁrocortisone arrow | Hyԁrocortisone biocodex | Hyԁrocortisone Upjohn;
- (GB) United Kingdom: Alkindi | Efcortelan | Efmody | Hydrocortison | Hydrocortison tab | Hyԁrocortisone | Hydrocortone | Plenadren | Solu cortef;
- (GR) Greece: Edicot | Hidroaltesona | Hyԁrocortisone | Hydrocortone | Lyo Cortin | Rolak;
- (GT) Guatemala: Hidrocortisona | Hidrocortisona succinato | Hydrobil in;
- (HK) Hong Kong: Cortef | Hyԁrocortisone | Hydrocortistab | Solu cortef | Uni Corsone;
- (HR) Croatia: Cortef | Plenadren | Solu cortef;
- (HU) Hungary: Alkindi | Cortef | Hydrocortison | Hydrocortison rompharm | Solu cortef;
- (ID) Indonesia: Fartison | Genisone | Hyԁrocortisone | Hyson | Lexacorton | Silacort | Solu cortef;
- (IE) Ireland: Cortef | Efcortelan soluble | Hydrocortone | Plenadren | Solu cortef;
- (IL) Israel: Solu cortef;
- (IN) India: Acucort | Alcort | Biosone | Celocort | Cipcorline | Cordro | Cort S | Cortireach | Cortiriv | Cortisafe | Cortisum | Cortopil | Efcorlin | H.s | Hekortin | Hisone | Hycorlin | Hycort | Hycoson | Hyԁrocortisone | Intacorlin | Ivcort | Labocort | Lycortin-s | Multicort | Novacort | Pilcort H | Primacort | Recort | Roscort | Succicort | Troycort | Unicort | Wosone | Wycort;
- (IQ) Iraq: Afatison;
- (IT) Italy: Alkindi | Cortidro | Cortop | Efmody | Flebocortid | Plenadren | Solu cortef;
- (JO) Jordan: Cortef | Cortistar | Hyԁrocortisone | Hydrosone;
- (JP) Japan: Cholidefeat | Cortef | Cortril | Excerate | Excerate taisho toyama iyak | Hyconate s | Hydrocorti.suc.ave | Hyԁrocortisone succinate na | Hydrocortone | Minasetyl | Saxizon | Scheroson f | Solu cortef | Succort;
- (KE) Kenya: Cardilan | Cort S | Cortidoz | Cortisol | Cortitab | Dawacort | Galcort | H cort | Hycorum | Labohydro | Lycortin-s | Medcort | Ocortin | Primacort | S hydro | Solu cortef | Umicort;
- (KR) Korea, Republic of: Corosone | Corticap | Cortisolu | Cortison | Cotico | Harond | Hiden | Hiroxon | Hydro | Hyrosine | Hyroson | Hyroxon | Hysone | Hytisone | Jr hyԁrocortisone | Rapison | Rotisone | Solu cortef | Whycort;
- (KW) Kuwait: Cortef | Solu cortef;
- (LB) Lebanon: Hyԁrocortisone | Solu cortef;
- (LT) Lithuania: Hc cort | Hisone | Hydrocortison | Hydrocortison Acetate | Hydrocortison acis | Hydrocortison rotexmedica | Hydrocortisonum sf | Hydrocortone | Solucortef | Sopolcort h;
- (LU) Luxembourg: Hyԁrocortisone | Hyԁrocortisone bepb | Solu cortef;
- (LV) Latvia: Hydrocortison | Hydrocortison Galen | Solucortef | Sopolcortholone h;
- (MA) Morocco: Flebocortid | Hyԁrocortisone roussel;
- (MX) Mexico: Cortef | Drosodin | Extaden | Fadol | Flebocortid | Flebonadrol | Flemex | Hidrocortis

gi | Hidrocortiso gi ly | Hidrocortiso.gi | Hidrocortisona | Inotid | Nositrol | Solhidrol | Sonatalis | Ticofran | Tisodank;

- (MY) Malaysia: Dhartisone | Hycort | Hyԁrocortisone | Solu cortef | Stricort-100;
- (NG) Nigeria: Acmecortisone | Chazmax hyԁrocortisone | Cortina | Darvinks hyԁrocortisone | Everdestiny hyԁrocortisone | Hydrokris | Longecut | Spansicort | Vinco hyԁrocortisone | Zonason hydrortisone sodium succinate;
- (NL) Netherlands: Acecort | Alkindi | Efmody | Hydrocortison | Hydrocortison activase | Hydrocortison teva | Hydrocortison tiofarma | Plenadren | Solu cortef;
- (NO) Norway: Alkindi | Cortef | Cortef euromedica | Efmody | Hidroaltesona | Hydrocortison jenapharm | Hyԁrocortisone accord | Hyԁrocortisone Alissa | Hydrocortone | Hydrokortison | Hydrokortison dak orifarm | Hydrokortison orion | Hydventia | Lilinorm | Plenadren | Solu cortef | Solucortef;
- (NZ) New Zealand: Solucortef;
- (PA) Panama: Hidrocortisona | Hidrocortisona klonal | Hybru | Solu cortef | Vesacort;
- (PE) Peru: Cortiston | Hidrocortisona | Hidrocortisona succinato sodico | Lycortin-s | Novocortil | Solu cortef | Solu cortef mix o;
- (PH) Philippines: Biocort | Corbal | Cortin | Cortis | Cortizol | Costeron H | Cylocortix | Eurocortef | Hycorlin | Hycort | Hycortil | Hydrobet | Hydrocortibas | Hydrotopic | Hydrovex | Hydrozone | Hyzonate | Lycortis | Pharmacort | Solu cortef | Sorvilor | Verdicort | Vitasone;
- (PK) Pakistan: Cartilan | Cortisol | Cortizone | Hicartif | Hycortisone | Hyԁrocort | Hyzonate | Solu cortef;
- (PL) Poland: Alkindi | Corhydron | Cortef | Hydrocortison | Hydrocortison Vuab | Hydrocortisonum | Hydrocortisonum aceticum | Hydrocortisonum jelfa | Hydrocortisonum sf | Hydroctin | Solucortef;
- (PR) Puerto Rico: A-hyԁrocort | Alkindi sprinkle | Cortef | Hyԁrocort sodium succinate | Hyԁrocortisone | Solu cortef;
- (PT) Portugal: Hidrocortisona color | Hidrocortisona generis | Hidrocortisona roussel | Hydrocortone | Rapicort | Solu cortef;
- (PY) Paraguay: Betacorten | Hidrocortisona | Hidrocortisona dallas | Hidrocortisona eticos | Hidrocortisona fda | Hidrocortisona fusa | Hidrocortisona imedic | Hidrocortisona klonal | Hidrocortisona lasca | Hidrocortisona nortelab | Hidrocortisona northia | Hidrocortisona prosalud | Hidrocortisona richmond | Zonac;
- (QA) Qatar: Cortef | Genkort | Hidrozon | Hydrosone | Solu-Cortef | UC-Cort | Zycort;
- (RO) Romania: Cortef | Flebocortid | Hidrocortizon HF | Hidrocortizon rompharm | Hydrocortison galepharm | Hyԁrocortisone roussel | Hyԁrocortisone succinat sodic | Hyԁrocortisone succinat sodic eipico;
- (RU) Russian Federation: Cortef | Hydrocortison | Solo cortef novamedica | Solu cortef;
- (SA) Saudi Arabia: Ikotref | Solu cortef;
- (SE) Sweden: Alkindi | Efmody | Hyԁrocortisone activase | Hydrokortison ebb | Hydrokortison nycomed | Hydrokortison Orifarm | Hydrokortison orion | Hydrokortison strides | Plenadren | Solu cortef;
- (SG) Singapore: Hyԁrocortisone | Solu cortef;
- (SI) Slovenia: Alkindi | Cortef | Hidrokortizon Altamedics | Hydrocortison | Hydrocortison hoechst | Hydrocortone | Solu cortef;
- (SK) Slovakia: Cortef | Hydrocortison | Hydrocortison Vuab | Hyԁrocortisone medopharm | Solucortef;
- (SL) Sierra Leone: Hycortin;
- (SR) Suriname: Hydro adreson | Hydrocortison | Solu cortef;
- (SV) El Salvador: Hidrocortisona succinato sodico;
- (TH) Thailand: Hydasone | Hydracort | Hydro-adreson aquosum | Hysone;
- (TN) Tunisia: Cortef | Hyԁrocortisone | Hyԁrocortisone Medis | Hyԁrocortisone panpharma | Steza | Suranil;
- (TR) Turkey: Genkort | Hidrozon | Hydrocortison;
- (TW) Taiwan: Efcortelan | Hyԁrocortisone | Hyԁrocortisone cyh | Saxizon | Solu cortef;
- (UA) Ukraine: Hyԁrocortisone rompharm | Solucortef;
- (UG) Uganda: Cortiriv | Hycort | Hyԁrocort | Hydrosone | Primacort | S hydro | Succicort | Umicort;
- (UY) Uruguay: Coripen | Fridalit | Hidrocortisona;

- (VE) Venezuela, Bolivarian Republic of: Corticina | Cortivid | Cyclocortid | Hidrocolsona | Hidrocort | Hidrocortisona | Histasona | Silian | Solucortef | Wellsona;
- (VN) Viet Nam: Gimtafort | Huhajo | Valgesic | Vinphason;
- (ZA) South Africa: Cortaject | Covocort | Solu cortef;
- (ZM) Zambia: Cipcorlin | Dawacort | Hycorcin | Lycortin-s | Primacort | Solu cortef;
- (ZW) Zimbabwe: Primacort | Solu cortef

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